

Praneeth Reddy Gandra

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EDUCATION

IIIT Tiruchirapalli

B.Tech, Computer Science and Engineering

2023 – 2027

Trichy, Tamil Nadu

- **CGPA:** 8.86 / 10.0 | **Semester 5 GPA:** 9.27 / 10.0

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, SQL

AI/ML & LLMs: PyTorch, Hugging Face, Keras, Scikit-learn, LangChain, Unsloth, FAISS, PEFT, QLoRA, NLP, Deep Learning, Computer Vision, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs)

Backend & APIs: FastAPI, REST APIs, Node.js, Express.js

Frontend: HTML, CSS, JavaScript, TypeScript, React

Databases: MongoDB, PostgreSQL, SQL, NoSQL

Cloud & Dev Tools: Cloud Deployment, Git, GitHub, Linux, Streamlit

AI Coding Tools: Cursor, GitHub Copilot, ChatGPT, Claude, Gemini

Libraries: Pandas, NumPy, Matplotlib

PROJECTS

LLM Fine-Tuning for Financial Sentiment Analysis | *PyTorch, Unsloth, PEFT*

Feb. 2026 – Mar. 2026

- Conducted an experimental research project exploring Parameter-Efficient Fine-Tuning (PEFT) and instruction-tuning using QLoRA
- Fine-tuned Metas Llama 3.2 (3B) Base model via Unsloth on a specialized Twitter financial dataset
- Successfully adapted a generative foundation model for strict sentiment classification, leveraging **4-bit quantization** and **Unsloth's optimized kernels** to drastically reduce GPU memory footprint and accelerate training

FDM 3D Printing Defect Detection | *PyTorch, Computer Vision, Transfer Learning*

Jan. 2026 – Feb. 2026

- Developed CNN-based models to detect defects in FDM 3D printing using computer vision techniques
- Evaluated models on two tasks: binary defect detection and 5-class defect classification (Cracking, Layer Shifting, Off Platform, Stringing, Warping)
- Benchmarked AlexNet, ResNet18, and InceptionV3 with ImageNet transfer learning, achieving **99.45%** binary accuracy and **99.22%** multi-class accuracy

Study Assistant Hub | *Python, LangChain, FAISS, OpenAI GPT, Streamlit*

Nov. 2025 – Dec. 2025

- Architected an AI-powered educational platform integrating a conversational chatbot, quiz generator, and flashcards ([GitHub](#))
- Implemented Retrieval-Augmented Generation (RAG) using FAISS for vector similarity search, enabling context-aware document querying and dynamic topic summarization

SMS Spam Detection | *NLP, Ensemble Methods*

Dec. 2024 – Jan. 2025

- Developed an NLP-based machine learning pipeline utilizing TF-IDF vectorization to classify SMS messages as spam or ham
- Trained and evaluated Multinomial Naïve Bayes, SVM, KNN, and Random Forest models, ultimately building a hard-voting ensemble classifier that achieved over 98.5% accuracy

Credit Card Default Prediction | *Machine Learning, Predictive Analytics*

Dec. 2024

- Built predictive analytics models to evaluate credit card default risk for clients in Taiwan using extensive financial history
- Benchmarked Logistic Regression, Decision Trees, SVM, Random Forest, Gradient Boosting, and XGBoost to identify the optimal predictor while addressing severe class imbalance in the dataset

RELEVANT COURSEWORK

AI & Mathematics: Artificial Intelligence, Introduction to LLMs, Linear Algebra, Probability & Statistics, Data Mining & Data Warehousing, Operations Research

Core CS: Data Structures, Design & Analysis of Algorithms, Computer Organisation & Architecture, Object-Oriented Programming, Compiler Design, Automata & Formal Languages, Discrete Structures

Systems & Networks: Operating Systems, Data Communication & Computer Networks, Database Management Systems, Software Engineering

ACHIEVEMENTS

- 1st Place Hackathon | IIT Trichy2025
- Awarded 1st place among peer teams for conceptualizing and developing an AI-powered study assistant within a strict 72-hour timeframe
 - Demonstrated rapid prototyping skills by integrating NLP tools to optimize study schedules and personalize learning

CERTIFICATIONS

- Supervised Machine Learning | Coursera (Stanford/DeepLearning.AI)Jan. 2025
- Credential: coursera.org/verify/PMVZ9RCBY7RL
- Crash Course on Python | Coursera (Google)July 2024
- Credential: coursera.org/verify/XXWA74WABWKV